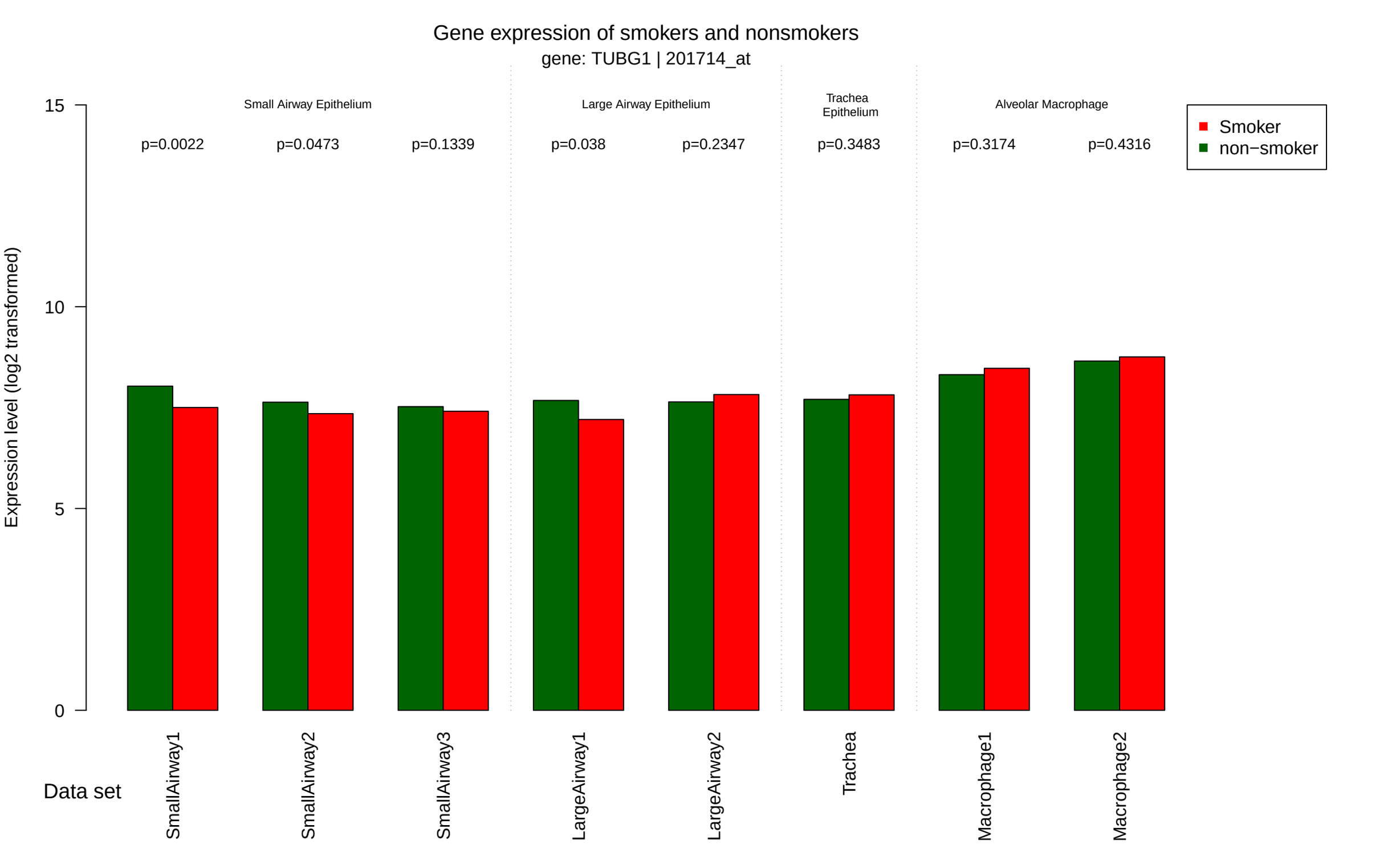
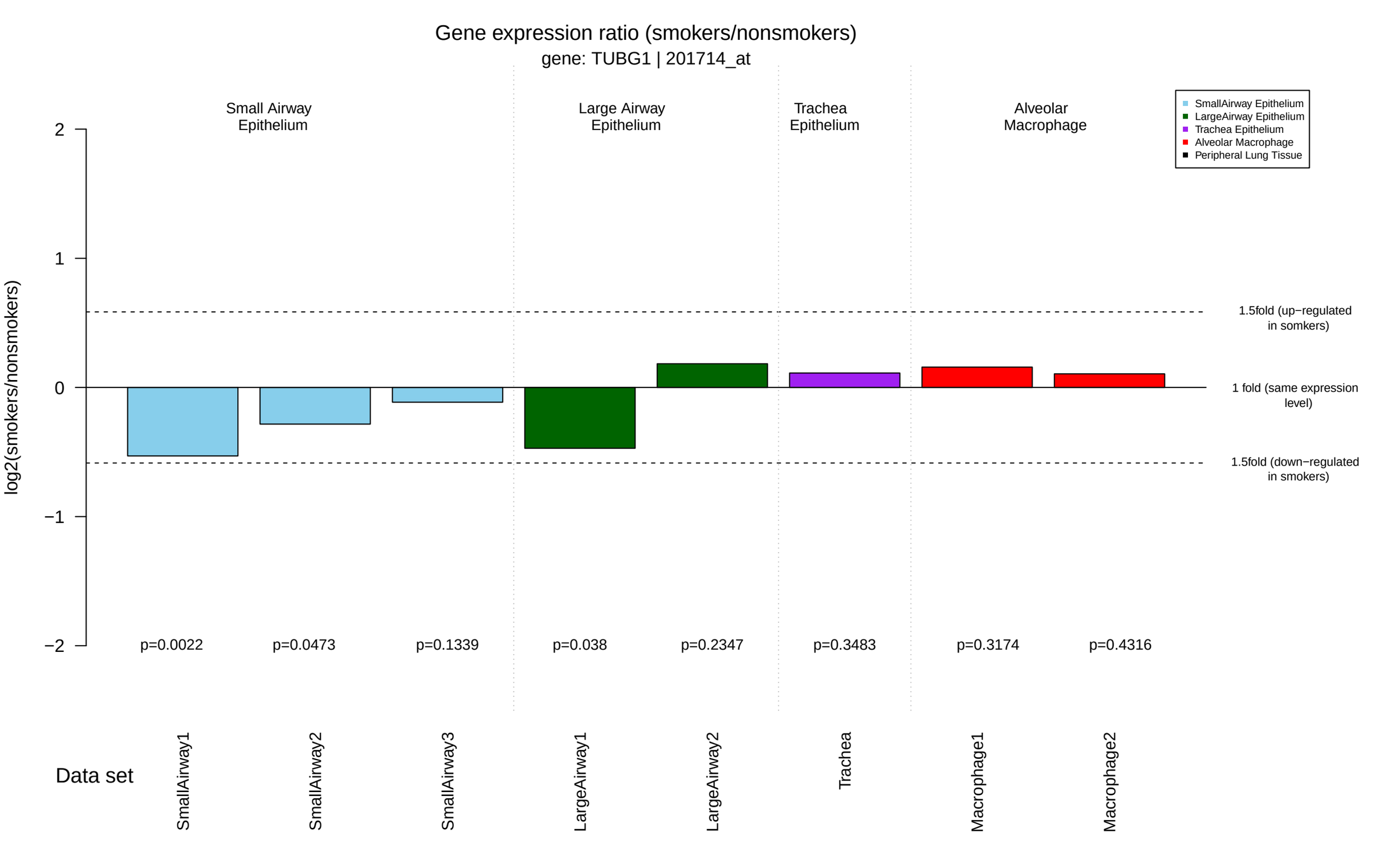
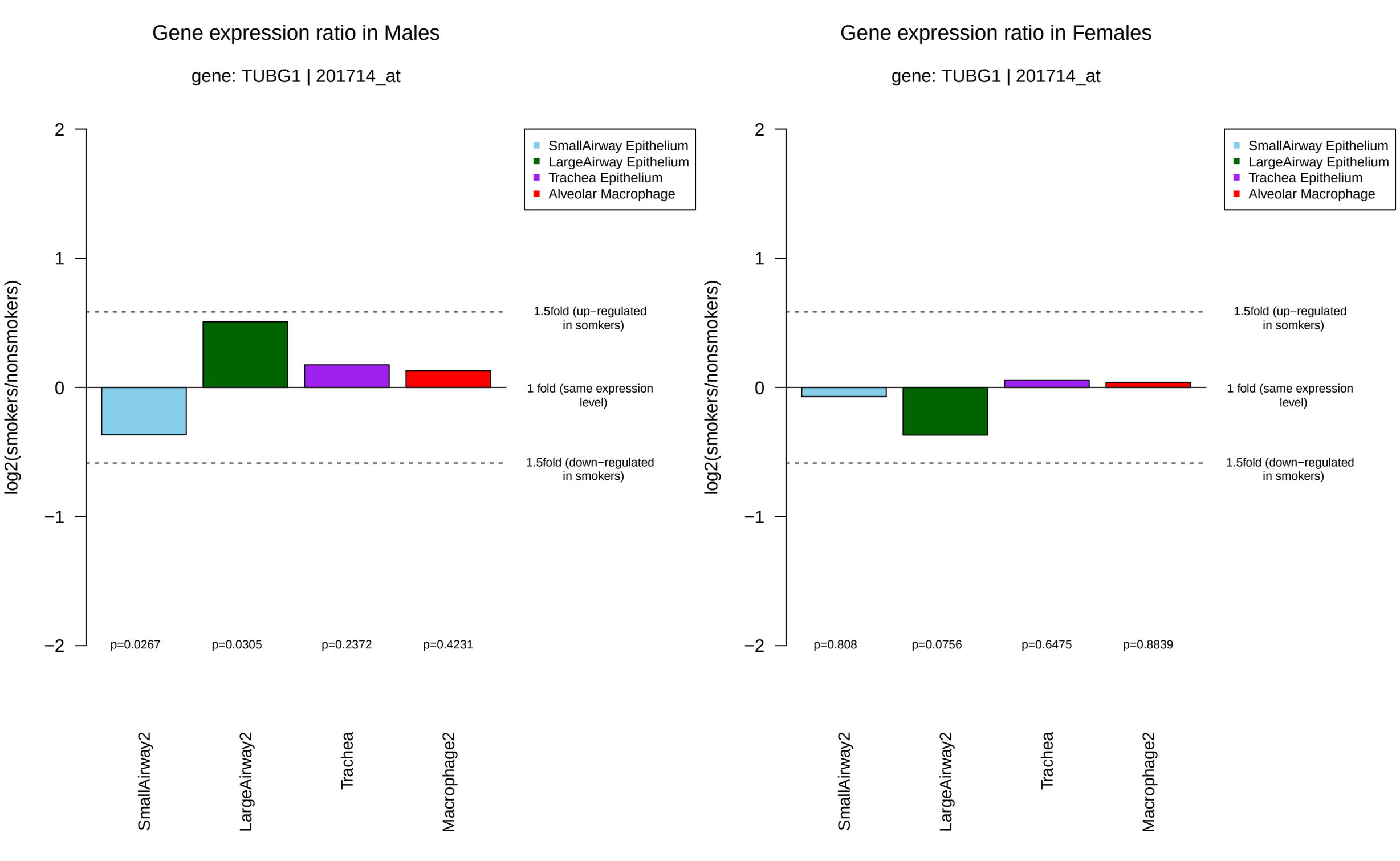
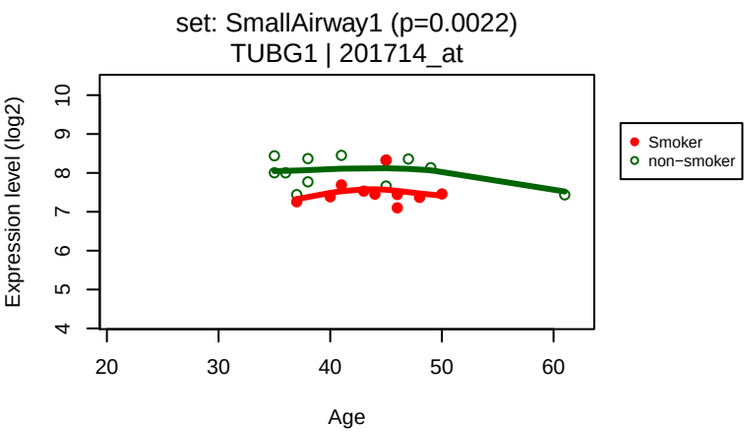
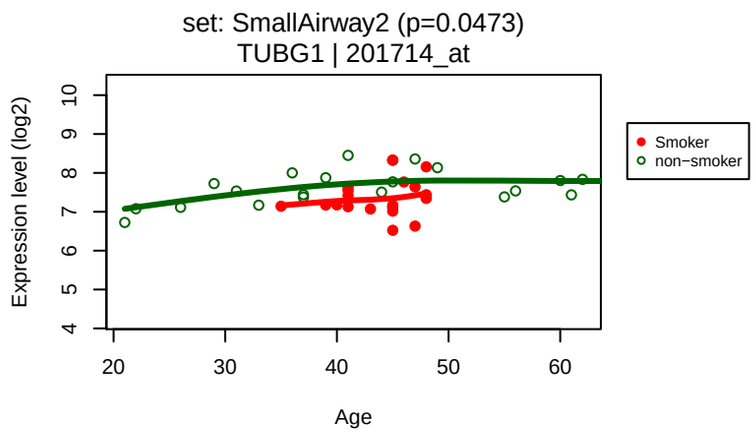
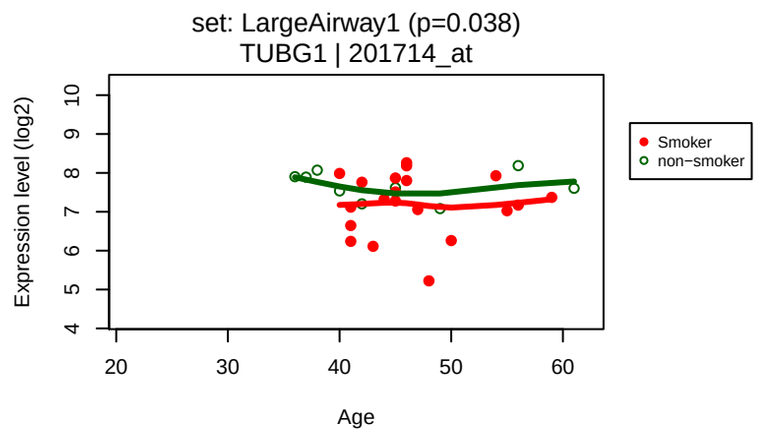

Gene expression of smokers and nonsmokers
gene: TUBG1 | 201714_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0022
p=0.0473
p=0.1339
p=0.038
p=0.2347
p=0.3483
p=0.3174
p=0.4316
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

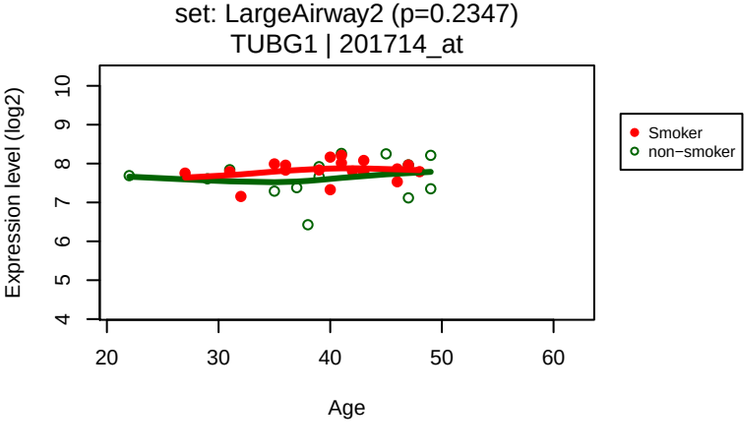

Gene expression ratio (smokers/nonsmokers)
gene: TUBG1 | 201714_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0022
p=0.0473
p=0.1339
p=0.038
p=0.2347
p=0.3483
p=0.3174
p=0.4316
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

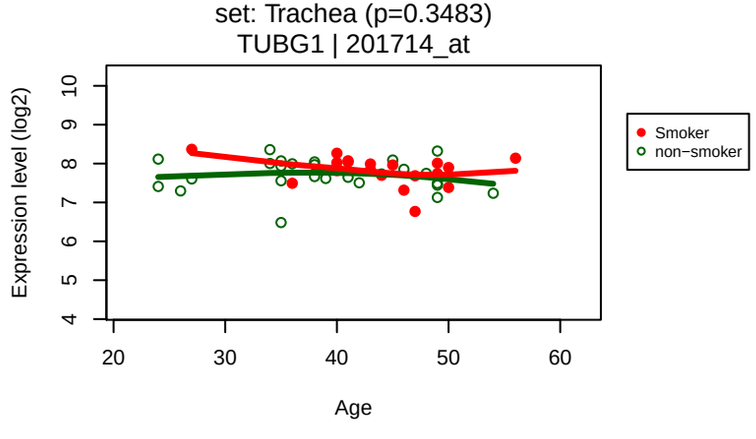

Gene expression ratio in Males
gene: TUBG1 | 201714_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0267
p=0.0305
p=0.2372
p=0.4231
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TUBG1 | 201714_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.808
p=0.0756
p=0.6475
p=0.8839
SmallAirway2
LargeAirway2
Trachea
Macrophage2

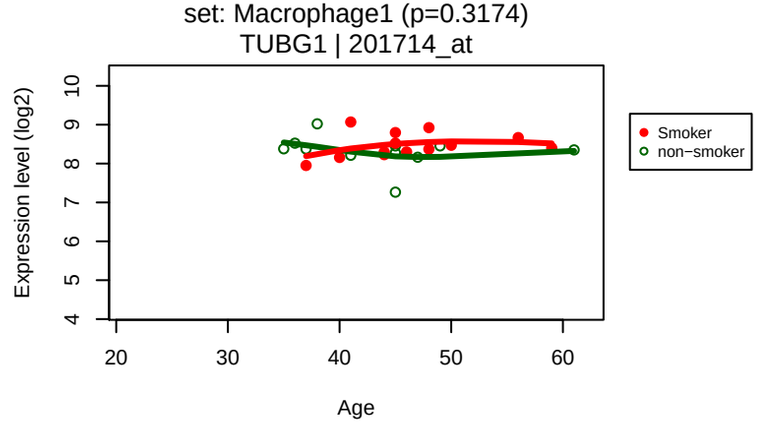

set: SmallAirway1 (p=0.0022)
TUBG1 | 201714_at
Expression level (log2)
Age
Smoker
non−smoker

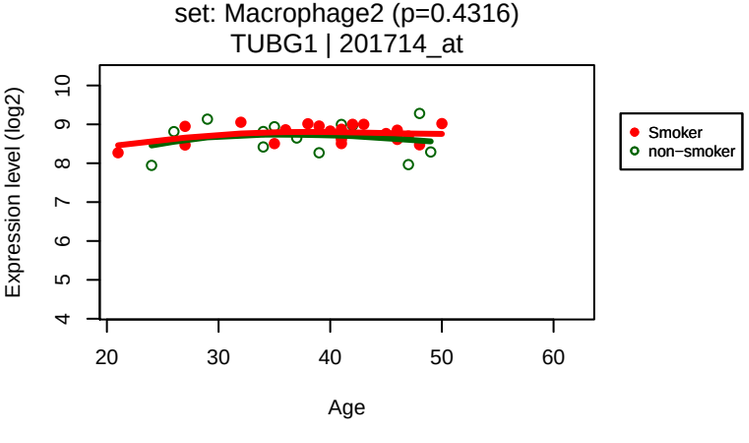

set: SmallAirway2 (p=0.0473)
TUBG1 | 201714_at
Expression level (log2)
Age
Smoker
non−smoker

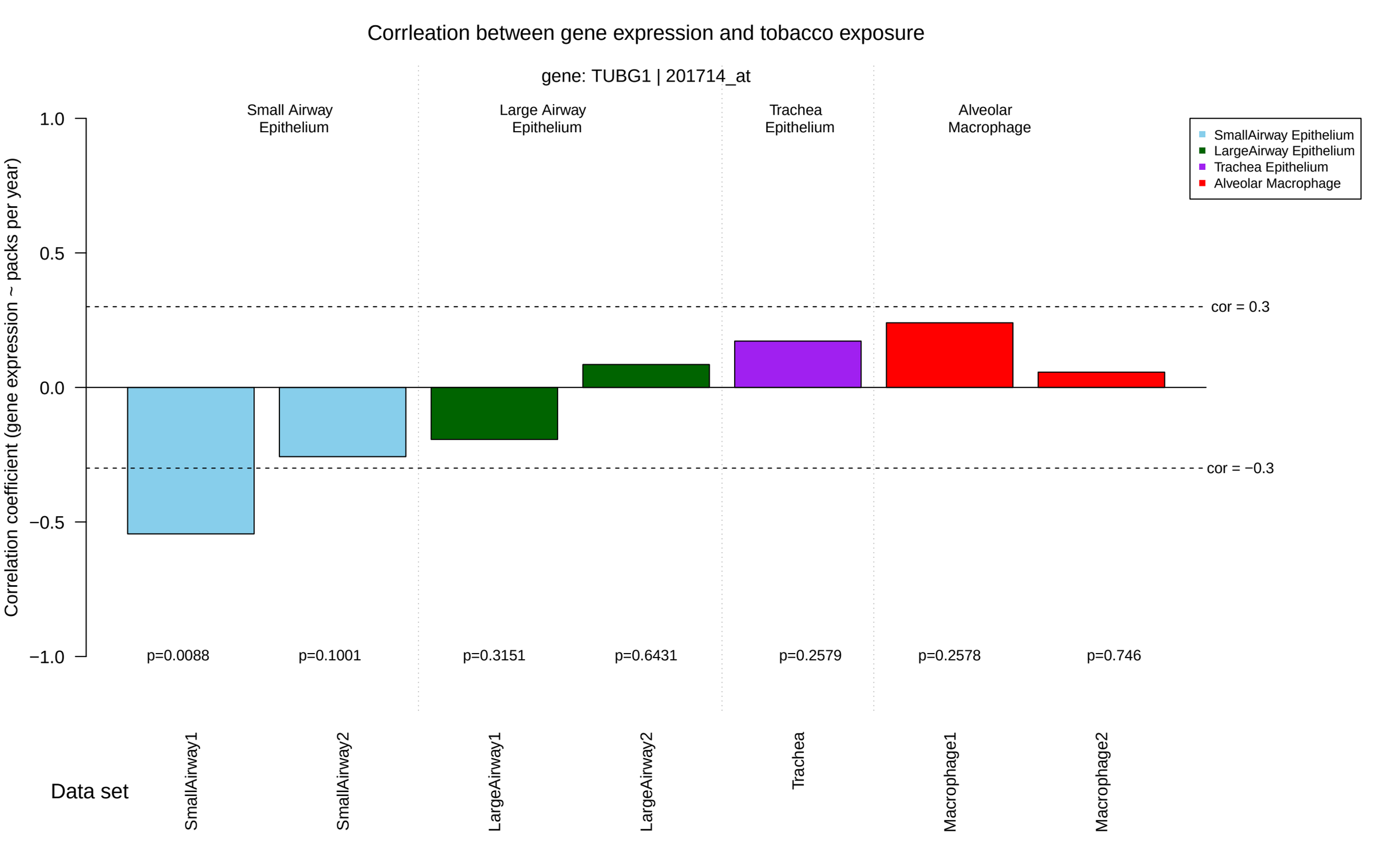

set: LargeAirway1 (p=0.038)
TUBG1 | 201714_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2347)
TUBG1 | 201714_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3483)
TUBG1 | 201714_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3174)
TUBG1 | 201714_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4316)
TUBG1 | 201714_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TUBG1 | 201714_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0088
p=0.1001
p=0.3151
p=0.6431
p=0.2579
p=0.2578
p=0.746
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2